

Connecting EXHALE to the Lower Atmosphere: Survey, Implementation, and Example Applications

1. Motivation and current state

EXHALE's lower boundary is the $\sim 1 \mu\text{bar}$ level: the input "Planet radius" is the *radius of that level* (user-guessed), the base temperature is fixed at $T_0 = T_{\text{eq}}$, the base density n_0 is an input (or pressure-anchored via `base_p_ubar`), and the gas is purely atomic H/He + trace metals at prescribed abundances. No molecules exist anywhere in the domain. The Wind-AE IC port carries a *simplified* molecular layer (μ -blend `molec_adjust` + bolometric heating/cooling, auto-shutoff when the atomic transition reaches the base).

This memo (i) surveys how the literature connects escape models to the lower atmosphere, (ii) documents the implementation in `src/modules/lower_atmosphere/` and `src/utis/run_lower.py`, (iii) applies it to the four planets modelled with EXHALE (HD 209458 b, HD 189733 b, WASP-121 b, WASP-52 b), and (iv) prescribes how the outputs are used and connected to EXHALE runs.

2. What the literature does at the interface

Approach	Lower atmosphere	Hands to the wind model at $\sim 1 \mu\text{bar}$	Cost / availability
Salz et al. (2016), TPCI	none — CLOUDY solves the whole column (molecules disabled)	n/a (single domain; base at $n = 10^{14} \text{ cm}^{-3}$, T_{eq})	$\sim 3 \times 10^5$ CPU-h for 18 planets; interface not public
Koskinen et al. (2022)	<i>analytic</i> : isothermal- T_{eq} hypsometric column from 1 bar, $\mu(p)$ from chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ fits (Visscher et al. 2006)	base <i>radius</i> $r_0(1 \mu\text{bar})$, $T_0=T_{\text{eq}}$, $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$; base v from mass conservation	trivial (closed-form; all equations in the paper)
Huang et al. (2017)	<i>analytic spacer</i> : isothermal $\mu=2.3$ column, 1 bar $\rightarrow 10 \mu\text{bar}$	base radius of the atomic layer; $\text{Ly}\alpha$ absorbing bottom boundary (H_2 accidental resonances, $\tau=1$ at $N_{\text{H}_2} \approx 10^{14} \text{ cm}^{-2}$)	trivial
Taylor et al. (2025, 2026)	full Lavvas & Arfaux (2021) model (RC temperature + ~ 150 -species kinetics of Lavvas et al. 2014 + haze microphysics; $10^3 \rightarrow 10^{-6}$ bar)	$T(1 \mu\text{bar})$, altitude of the $1 \mu\text{bar}$ level, composition (with a <i>tolerated discontinuity</i>), K_{zz}	in-house, not public ; coupling is a manual splice, iterated $\sim 3 \times$

Key quantitative facts:

- **Base insensitivity for strong winds** (Salz et al. 2016, §3.5–3.6): $T_{\text{base}} \pm 40\% \Rightarrow \dot{M} \pm 5\text{--}10\%$ (linear); base density $10^{14} \rightarrow 10^{12} \text{ cm}^{-3} \Rightarrow \dot{M} \lesssim 40\%$ lower. Their error budget: base density $\pm 50\%$, base $T \pm 10\%$ — both dwarfed by the XUV uncertainty ($\times 3$). $\Rightarrow$ EXHALE’s fixed- T_{eq} base is already defensible for hot Jupiters.
- **Where the fixed base breaks:** (i) planets near radiative equilibrium (stable thermospheres); (ii) cool/sub-Neptune planets where H_2 survives into the wind — Salz’s GJ 1214 b test: H_2 10–25% throughout, H^- a significant coolant, $\dot{M} - 15\%$; Taylor et al. (2026): adding H_2 halves the He 10830 signal (14% $\rightarrow$ 7%); Koskinen et al. (2022) hot Uranus: base $q_{\text{H}_2} \approx 0.84$ and H_3^+ IR emission is the dominant radiative coolant below $\sim 2.3 R_p$.
- **What a photochemical lower model uniquely provides** (Lavvas et al. 2014; Lavvas & Arfaux 2021): the $\text{H}_2 \rightarrow \text{H}$ transition can sit exactly at 1 μbar and is temperature-sensitive there; photochemistry (OH-catalysed H_2 destruction) dissociates H_2 even at 500–1000 K; *atomic-metal release profiles* (Na, K, Mg, Ca, Fe, Al, Si return to atomic form below $\sim 10^{-3}$ bar) — the physically-motivated base abundances for EXHALE’s trace metals; photoionization-dominated electron density ($\sim 10^8 \text{ cm}^{-3}$, $100\times$ Saha); radical/haze heating of $\pm 100\text{--}400$ K right at the handoff level.
- **The published coupling is loose, not monolithic:** Lavvas et al. (2014) take $T(p < 1 \mu\text{bar})$ from the Koskinen thermosphere and use “species < 3 amu escape at the wind velocity” as their upper BC; Taylor et al. tolerate a composition discontinuity at 1 μbar . Only TPCI runs a tight coupling at every timestep (and it disables molecules).

3. The three-tier plan

Tier 1 — analytic lower column (Koskinen-2022 style): derive the 1- μbar base radius and base $\text{H}_2/\text{H}/\text{He}$ from a closed-form column. **Tier 2** — molecular extension of EXHALE (H_2 , H_2^+ , H_3^+ , HeH^+ chemistry + H_3^+ IR cooling), required only for warm Neptunes / sub-Neptunes; **Tier 2a** — an interim EOS-only molecular-base correction. **Tier 3** — consume a real lower-atmosphere model (photochemistry + radiative–convective $T(p)$) through a file interface; the open-source VULCAN (Tsai et al. 2017, 2021) + HELIOS (Malik et al. 2017) stack is the practical replacement for the non-public Lavvas & Arfaux code, and a TPCI-style full coupling is rejected (cost, availability, and demonstrated base insensitivity). Which science case needs which tier: hot Jupiters $\rightarrow$ Tier 1 only; He 10830/ $\text{H}\alpha$ work $\rightarrow$ Tier 1; metal transmission lines $\rightarrow$ Tier 3 (atomic-metal release abundances); warm Neptunes / sub-Neptunes $\rightarrow$ Tier 2 + 1.

4. Implementation

All Fortran lives in `src/modules/lower_atmosphere/`; the Python driver in `src/utils/`. Everything is **opt-in / default off** — with no keys and no `base.inp`, standard runs are unchanged by the lower-atmosphere code (the molecular chemistry off-switch reproduces the atomic systems byte-for-byte; the absolute production baseline itself moved with the 2026-07-23 defaults — staged secondary ionization, `He_rec_coupling`, He–H charge exchange).

4.1 Tier 1: lower_column.f90

Equations. The isothermal hypsometric relation is integrated from the 1-bar level (radius $R_{1\text{bar}}$, in practice the broadband transit radius) to the base pressure $p_0 = 1 \mu\text{bar}$,

$$\frac{d \ln p}{dr} = -\frac{\mu(p, T) m_{\text{H}} g(r)}{k_{\text{B}} T}, \quad g(r) = \frac{GM_{\text{p}}}{r^2}, \quad T = T_{\text{eq}}, \quad (1)$$

with a fixed-step 4th-order Runge–Kutta in $x = \ln p$ (2000 steps; the integrand is smooth). The mean molecular weight follows the chemical-equilibrium H_2/H partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022, Eq. 11),

$$q_{\text{H}_2} \simeq \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p_{[\text{bar}]} + 6.2645. \quad (2)$$

An implementation subtlety uncovered during validation: Eq. (2) returns the *full-mixture* volume mixing ratio $q_{\text{H}_2} = n_{\text{H}_2}/(n_{\text{H}_2} + n_{\text{H}} + n_{\text{He}})$ (a protosolar He abundance is baked into the fit), so the fraction x_2 of H *nuclei* bound in H_2 must be recovered as

$$x_2 = \frac{2q_{\text{H}_2}(1 + f_{\text{He}})}{1 + q_{\text{H}_2}} \leq 1, \quad \mu = \frac{1 + 4f_{\text{He}}}{(1 - x_2) + x_2/2 + f_{\text{He}}} \text{ [amu]}, \quad (3)$$

with $f_{\text{He}} = n_{\text{He}}/n_{\text{H(nuclei)}}$ the elemental ratio (EXHALE’s HeH). Equation (3) was verified against the Koskinen et al. (2022) Model A base composition ($q_{\text{H}_2} = 0.84$, $q_{\text{H}} = 0.026 \Rightarrow x_2 = 0.985$); the naive H-subsystem inversion $x_2 = 2q/(1 + q)$ is *wrong* and misses the gate.

Interface. `lower_column_solve(Mp, R_deep, T, fhe, p_deep, p_base, r_base, qh2_b, qh_b, qhe_b, mu_b, r_base_atomic)` (cgs; pressures in bar). Because chemical equilibrium *underestimates* H dissociation at $T_{\text{eq}} \approx 1000\text{--}2000$ K (photochemistry ignored; Moses et al. 2011; Koskinen et al. 2013a), the routine also integrates the same column with the fully atomic $\mu = (1 + 4f_{\text{He}})/(1 + f_{\text{He}})$ and returns `r_base_atomic`: the two radii *bracket* the truth. The public helper `q_h2_equilibrium(p, T)` exposes Eq. (2) (NaN-safe, clamped to $[0, 1]$).

EXHALE key. Lower column: `<R_1bar in R_J>` in `input.inp` triggers a startup report: the derived $r_0(1 \mu\text{bar})$ (equilibrium + atomic bracket), the input “Planet radius” for comparison, and the base $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}/\mu$, plus a warning when the equilibrium base is strongly molecular. *Report only — nothing is overridden* (overriding is Tier 3’s job, §4.4).

Validation gate (passed). Koskinen et al. (2022) Model A (Uranus-mass planet, $M = 8.6813 \times 10^{28}$ g, $R_{1\text{bar}} = 2.5559 \times 10^9$ cm, Sun-like star at 0.05 au, $T_{\text{eq}} = 1140$ K, $f_{\text{He}} = 0.0793$):

	$r_0/R_{1\text{bar}}$	$q_{\text{H}_2}(\text{base})$	$q_{\text{H}}(\text{base})$
this module	1.343	0.838	0.027
Koskinen et al. (2022)	1.34	≈ 0.84	≈ 0.026

Approximations (documented in the source): isothermal column at T_{eq} ; spherical gravity (the Roche-equipotential correction matters only near RLO); chemical equilibrium only.

4.2 Tier 2a: the Molecular base key

Molecular base: True applies an *EOS-only* correction at startup: the H nuclei bound into H₂ at (1 μbar, T₀) per Eq. (2)–(3) are removed from the base particle budget,

$$\text{ntot_bc} \rightarrow \text{ntot_bc} - \frac{x_2/2}{1 + f_{\text{He}}}, \quad (4)$$

which lowers the base pressure ($p_{\text{base}} = \text{ntot_bc} n_0 k_B T_0$) and raises the base mean molecular weight, i.e. a heavier, more compact base — the leading-order effect of a molecular base on the hydrodynamics. The species arrays remain atomic (H₂ chemistry is Tier-2 proper), so this is a *crude, clearly-flagged* approximation. Example: HD 209458 b, $q_{\text{H}_2}(1 \mu\text{bar}, 1450 \text{ K}) = 0.831 \Rightarrow \text{ntot_bc} 1.0 \rightarrow 0.546$.

4.3 Tier 2 foundation: h3p_cooling.f90 and mol_rates.f90

H₃⁺ infrared cooling. Optically thin cooling per molecule from Miller et al. (2013, Table 5, $z(T)$ fits, archived as references/Miller_2013_JPCA_117_9770.pdf):

$$\ln E(\text{H}_3^+, T) = \sum_n C_n T^n \quad [E \text{ in W molecule}^{-1} \text{ sr}^{-1}], \quad (5)$$

piecewise over 30–300, 300–800, 800–1800, and 1800–5000 K (fit errors < 0.1% over 300–5000 K), held frozen above 5000 K (H₃⁺ is destroyed well below). The volumetric rate is

$$\Lambda_{\text{H}_3^+} = n_{\text{H}_3^+} 4\pi E(T) s_{\text{nonLTE}}(T, n_{\text{H}_2}) \times 10^7 \quad [\text{erg s}^{-1} \text{ cm}^{-3}], \quad (6)$$

where s_{nonLTE} is the departure factor of Miller et al. (2013, Table 6; their Oka & Epp detailed-balance approach), implemented as a bilinear interpolation in $(T, \log_{10} n_{\text{H}_2})$ over $T = 300\text{--}5000 \text{ K}$ and $n_{\text{H}_2} = 10^6\text{--}10^{14} \text{ cm}^{-3}$; $s \rightarrow 1$ (LTE) at high density, and the tabulated values are upper limits per the paper’s caveat on the proton-hopping rate. Both interpolators use the NaN-safe .not.(x>edge) bracketing adopted after the 2026-07-02 code review. **Validation:** the fits reproduce the paper’s Table-4 anchors at $T = 500/1000/2000/3000/5000 \text{ K}$ to <0.5%, and $s(1000 \text{ K}, 10^{10} \text{ cm}^{-3}) = 0.4955$, $s(2000 \text{ K}, 10^{12}) = 0.9499$ exactly.

Molecular reaction rates. mol_rates.f90 provides the full H₂/H₂⁺/H₃⁺/HeH⁺ thermochemical network of Koskinen et al. (2022, Table 1; transcribed and verified against the PDF page): recombination (R1–R2, Storey & Hummer 1995), Voronov (1997) collisional ionization (R3–R4), H₂⁺/H₃⁺/HeH⁺ dissociative recombination (R5–R7, R16; Larsson et al. 2008), the H₃⁺ production backbone H₂⁺ + H₂ → H₃⁺ + H (R8, 2×10^{-9}), thermal and electron-impact H₂ dissociation (R12, R14), three-body chemistry (R13, R15), the HeH⁺ chain (R17–R20), and H/He charge exchange (R21–R23). The stellar photo-rates P1–P5 (including H₂ dissociative and double photoionization) follow EXHALE’s existing PH_heat routines once H₂ cross sections are added. Baseline caveat inherited from the source: Lyman–Werner neutral photodissociation is not in the network (Koskinen et al.’s sensitivity test bounded its effect on $\dot{M}$ at $\times 1.4$).

Tier-2 core (implemented 2026-07-03). The coupled system System_HeH_mol.f90 solves 7–8 unknowns per cell (H⁺, He⁺, He⁺⁺, optionally He 2³S, plus the H-nucleus fractions bound in H₂, H₂⁺, H₃⁺, HeH⁺) as element-conserving

steady-state balances with `hybrd1`: the atomic rows use EXHALE's own rate arrays (so the molecule-free limit reproduces the atomic systems' solution) and the molecular channels are the Table-1 network via `mol_rates` (R21/R22 H–He charge exchange deliberately excluded to preserve the atomic limit). The H_2 photoionization cross section (Yan et al. 1998, Eqs. 17–19; verified against their Table 7 to $<0.2\%$) enters the opacity, the P_{H_2} photoionization integral, the photoelectric heating, and the absorbed-energy normalization through optional arguments of `PH_heat_HHe`; H_3^+ IR cooling enters the `cool` array. The EOS counts each molecule as one particle, each molecular ion's electron, and the correct masses (`calc_ne/ntot/rho`, `get_species_densities`). `f_sp` grows to 37 columns (output schema and Load-IC names extended). Key: Molecular chemistry: True (requires He; v1 excludes trace metals; the molecular base particle count is coupled on automatically — an atomic `ntot_bc` under a molecular base is physically inconsistent). **Gates passed** (numbers re-established 2026-07-16 after the deterministic chemical-equilibrium seeding of the molecular solve; the exact gate inputs are pinned at `lower_atmosphere_figs/data_g*/input.inp`): (0) molecules-off regression, molecular chemistry off reproduces the atomic systems; (1) HD 209458 b molecules-on: a fully molecular base ($x_{\text{H}_2} \simeq 0.996$) with a sharp $\text{H}_2 \rightarrow \text{H}$ front at $r = 1.020 R_p$ and an essentially atomic wind above it — the hot-Jupiter atomic base is now a *result*, not an assumption — while below the front the molecular base suppresses $\text{He } 2^3\text{S}$ by orders of magnitude ($\text{He } 2^3\text{S} + \text{H}_2$ Penning destruction, $\sim 10^4 \times$ at the front, rising back to the atomic value above $\sim 1.3 R_p$); (2) a hot-Uranus-like case ($0.0457 M_J$, $R_p = 0.49 R_J$, $T_{\text{eq}} = 1140 \text{ K}$, HD 209458 b orbit and spectrum): the front sits at $r = 1.156 R_p$ and H_3^+ is active below it (peak $\sim 7 \times 10^4 \text{ cm}^{-3}$ at $r \approx 1.04$). The $\text{H}_2 \rightarrow \text{H}$ fronts and molecular base composition are essentially unchanged from the 2026-07-16 gates (front $1.019 \rightarrow 1.020$, $1.166 \rightarrow 1.156$), confirming the dissociation-front result is robust to the 2026-07-23 production defaults; refreshed 2026-07-23 at a shared 12000-step relaxation-snapshot convention (both HD209 runs $\log \dot{M} = 10.58$; relaxation snapshots, not flux-flat converged). Figure 1 shows the gate runs.

He 2^3S + H_2 Penning ionization. When `Include_He23S?` is also on, the metastable couples to H_2 through Penning ionization $\text{He}(2^3\text{S}) + \text{H}_2 \rightarrow \text{He}(1^1\text{S}) + \text{H}_2^+ + e^-$ — the dominant $\text{He}(2^3\text{S})$ loss toward an H_2 -dominated base (García Muñoz 2025, A&A 698, A199, Fig. 4). Its rate coefficient `penning_HeI23S_H2` (`Cool_coeff.f90`) is an analytic fit to García Muñoz Table A.5 (Cohen & Lane 1977 cross sections), $k(T) = 5.3791 \times 10^{-12} T^{0.6760} e^{-695.21/T} \text{ cm}^3 \text{ s}^{-1}$, reproducing the tabulated 500–10000 K points to $\leq 0.13\%$. It enters `System_HeH_mol` as an H_2 loss / H_2^+ source / $\text{He}(2^3\text{S})$ sink and adds $(E[2^3\text{S}] - \text{IP}[\text{H}_2]) \approx 4.4 \text{ eV}$ of electron heating; the minor associative $\text{H} + \text{HeH}^+$ branch ($\sim 10\%$) is folded into the Penning channel.

The `_adv` post-process is atomic-only, so the Penning- H_2 term does not appear there. The advection-corrected reconstruction in `post_process_adv.f90` (header, lines 4–13) treats the gas as H/He + trace metals and excludes the molecular species entirely: the molecular densities are never passed in or re-solved, and its `calc_ne/calc_ntot` calls omit the `nmol` argument. Consequences for a molecular run: (i) the `_adv` n_{tot}/n_e omit the neutral- H_2 particle count and the molecular-ion electrons; (ii) the $\text{He}(2^3\text{S}) + \text{H}_2$ Penning sink cannot act in the `_adv` solve (no n_{H_2} there), so the `_adv` outputs — and the transit module that reads them — carry the *equilibrium* molecular-base triplet suppression ($39 \times$ at the HD 209458 b base, $430 \times$ on the hot-Uranus case) without an advection correction. This is acceptable where `_adv` is physically meaningful (the atomic/ionized escape flow above the $\text{H}_2 \rightarrow \text{H}$ front); below the front the atomic `_adv`

approximation is outside its validity domain anyway. A molecular-aware `_adv` is a separate extension, naturally paired with the open “molecular advection” item in the Tier-2 remaining list.

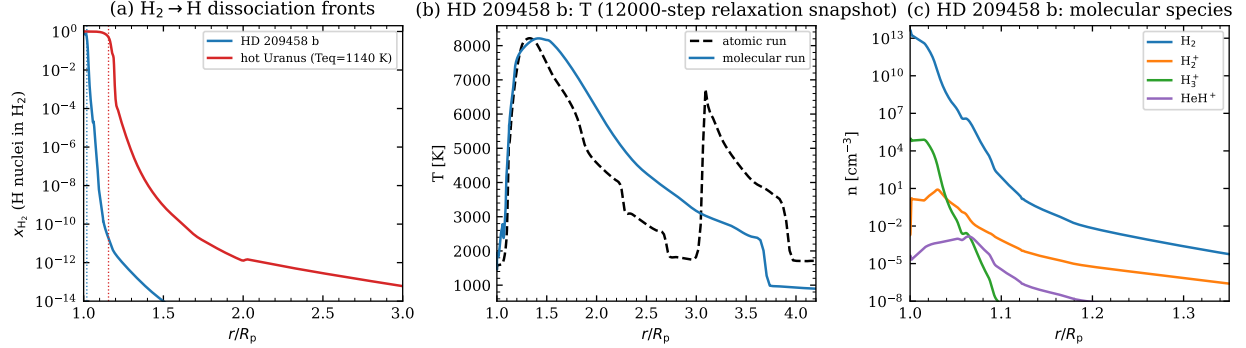


Figure 1: Tier-2 molecular gates (12000-step relaxation snapshots). (a) The H nuclei fraction bound in H₂: on HD 209458 b the base is fully molecular and a sharp H₂ → H front sits at $r = 1.020R_p$ (dotted); on the cooler, low-gravity hot Uranus the front sits at $1.156R_p$. The cell-to-cell flip-flop just above the front marks the chemically bistable zone of *equilibrium* H₂ chemistry — kinetics+advection smooth it (the documented Tier-2 caveat). (b) The molecular run’s temperature agrees with the atomic run away from the (still-relaxing) front region. (c) Molecular species on HD 209458 b: H₃⁺ (green) and the molecular ions are confined below the front, as expected for a hot Jupiter.

Remaining Tier-2 work (open). Molecular *advection* (the local chemical-equilibrium treatment cannot reproduce Koskinen et al.’s advective replenishment of high-altitude H₂ on cool Neptunes); Lyman–Werner photodissociation ($\lesssim \times 1.4$ on $\dot{M}$ per their test); the dissociative/double photoionization channels (P4/P5); the 4.48 eV dissociation energy sink; a diatomic γ ; the metals+ molecules merge; and the GJ 1214 b He-halving validation (Taylor et al. 2026) with a full M-dwarf setup.

4.4 Tier 3: base.inp and run_lower.py

The handoff file. An optional `base.inp` in the run directory is read by `input_read` *before* the derived constants, so its overrides propagate consistently (v_0 , b_0 , grid, normalization):

key	unit	overrides	used by
T_base	K	T_0 (base temperature)	thermal BC, EOS scales
r_base	R_J	“Planet radius” ($= r_0$ of $1 \mu\text{bar}$)	grid, gravity, areas
HeH_base	–	He/H number ratio	composition, He diffusion
Kzz_base	cm ² /s	he_kzz	He_diffusion homopause
q_H2_base	–	q_h2_base (q_{H_2} at the base)	Molecular base particle count
p_base	bar	p_base_bar	level of the handoff; the q_{H_2} fit

Every override is echoed to stdout; an absent file is a strict no-op. Unknown keys and # comments are ignored, so a richer lower-atmosphere model may carry extra metadata in the same file.

The driver. `src/utls/run_lower.py <run_dir> --r1bar <R_J> [--guillot] [--tint T] [--kzz K]` reads M_p , T_{eq} , He/H from the run directory's `input.inp`, integrates the same column as §4.1 in Python, and writes `base.inp`. Two temperature models: isothermal T_{eq} (default; identical to the Fortran Tier-1 column) and the Guillot (2010) semi-grey global-average profile

$$T^4(\tau) = \frac{3T_{int}^4}{4} \left(\frac{2}{3} + \tau \right) + \frac{3T_{eq}^4}{4} \left(\frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left[\frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}} \right] e^{-\gamma\tau\sqrt{3}} \right), \quad \tau = \kappa_{IR} p / g, \quad (7)$$

with standard hot-Jupiter defaults ($\kappa_{IR} = 10^{-2} \text{ cm}^2 \text{ g}^{-1}$, $\gamma = 0.4$, $T_{int} = 100 \text{ K}$). The file records the base $q_{H_2}/q_H/q_{He}$ as comments and prints a molecular-base warning when $q_{H_2} > 0.3$. The upgrade path — running VULCAN for the composition and a real RC model for $T(p)$ — writes the *same* `base.inp` keys, so nothing on the EXHALE side changes.

VULCAN upgrade implemented and demonstrated (2026-07-03). The public VULCAN code (cloned to `./VULCAN`; FastChem compiled) was run to steady state for HD 189733 b (SNCHO-2025 photochemical network, 2356 steps) and converted with the new `src/utls/vulcan_to_base.py` (`.vul` → `base.inp`: photochemical $q_{H_2}/q_H/q_{He}$ at $1 \mu\text{bar}$, hypsometric r_{base} with VULCAN's own $\mu(p)$ and $T(p)$, molecular mixing ratios as comments). **Result:** at $1 \mu\text{bar}$ the photochemical base is $q_{H_2} = 0.63$, $q_H = 0.23$ — *photochemistry dissociates $\sim 11\times$ more H than the chemical-equilibrium column* ($q_H = 0.020$), directly quantifying the Tier-1 caveat (Moses 2011; Koskinen 2013a); $r_{base} = 1.168$ vs the analytic $1.174 R_J$; $T_{base} = 863 \text{ K}$ (the Moses11 $T(p)$, cooler than the isothermal $T_{eq} = 1183 \text{ K}$). The full chain `VULCAN` → `converter` → `base.inp` → EXHALE startup override was exercised end-to-end. Caveat: VULCAN's networks are H/C/N/O/(S) — the atomic-metal release (Na/Mg/Ca/Fe) has no public analogue and remains with the Lavvas collaboration path; `metals.inp` stays user-supplied. Figure 2 shows the converged composition against the equilibrium H_2 track.

The photochemical q_{H_2} now reaches the solve (2026-08-10). Until then the converter recorded q_{H_2} as a comment while the Molecular base correction took its own q_{H_2} from the chemical-equilibrium fit — the very approximation the pre-step exists to replace. `base.inp` now carries two further read keys, `q_H2_base` (the photochemical H_2 volume mixing ratio at the base) and `p_base` (the level it refers to, at which the fit is evaluated so that fit and handoff always describe the same level). Both converters write them, and the base particle count is built in one place (`comp_ntot_bc` in `composition.f90`) from the photochemical value when it is present and from the fit otherwise, so a `base.inp` without the key behaves exactly as before. For HD 189733 b the VULCAN value $q_{H_2} = 0.6181$ against the fit's 0.8384 moves the subtraction from `ntot_bc` by 0.074 per nucleus ($\sim 7\%$), which the base density inherits in Base BC: pressure mode. Design and validation: `docs/base_composition_handoff_plan.md`.

5. Example applications: HD and WASP planets

The approach was applied to the four planets modelled with EXHALE (`examples/13_lower_atmosphere/`; $R_{1bar} =$ broadband transit radius; T_{eq} , M_p , He/H from each run's `input.inp`).

Interpretation (per planet):

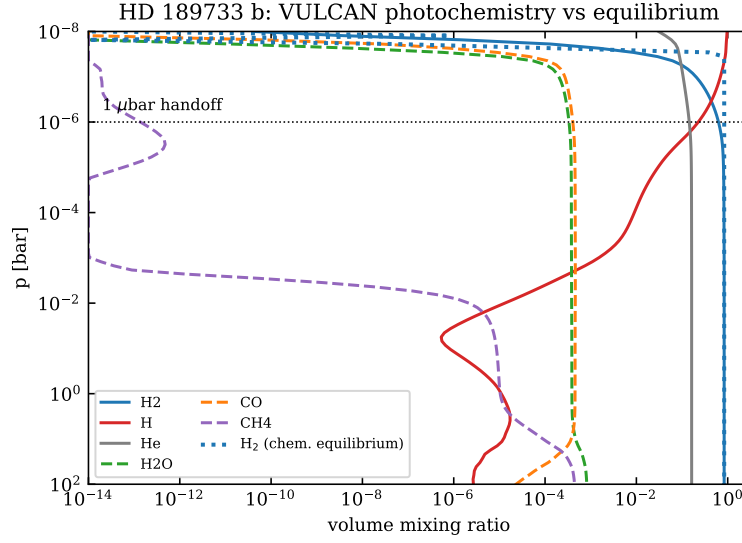


Figure 2: HD 189733 b lower/middle atmosphere from the converged VULCAN run (solid/dashed: H_2 , H , He , H_2O , CO , CH_4) compared with the chemical-equilibrium H_2 mixing ratio evaluated along the same $T(p)$ (dotted). Above $\sim 10^{-4}$ bar photochemistry dissociates H_2 far beyond equilibrium; at the $1 \mu\text{bar}$ handoff (horizontal dotted line) $q_{\text{H}} = 0.23$ versus the equilibrium 0.020 — the $\sim 11\times$ underestimate quantified.

	M_{p}	T_{eq}	He/H	$R_{1\text{bar}}$	input R_0	$r_0(1 \mu\text{bar}) [R_{\text{J}}]$		r_0^{Gui}	q_{H_2}	μ_{base}
planet	$[M_{\text{J}}]$	$[\text{K}]$		$[R_{\text{J}}]$	$[R_{\text{J}}]$	chem.eq.	atomic	$[R_{\text{J}}]$	(base)	[amu]
HD 209458 b	0.720	1450	0.0833	1.36	1.401	1.472	1.582	1.466	0.831	2.25
HD 189733 b	1.237	1183	0.0833	1.138	1.193	1.174	1.205	1.171	0.838	2.26
WASP-121 b	1.182	2358	0.0851	1.753	2.2075	2.010	2.133	1.966	0.005	1.24
WASP-52 b	0.460	1304	0.0204	1.27	1.270	1.437	1.615	1.428	0.837	1.95

Isothermal- T_{eq} column (chem. equilibrium + fully-atomic bracket) and the Guillot variant r_0^{Gui} ($T_{\text{base}}^{\text{Gui}} = 1313, 1072, 2136, 1181 \text{ K}$ respectively). “input R_0 ” is the base radius currently used in the production runs.

- **HD 189733 b — consistent.** The production base radius ($1.193 R_{\text{J}}$) falls *inside* the derived bracket $[1.174, 1.205]$: the adopted setup is validated by the column, not merely assumed.
- **HD 209458 b — input $\sim 5\%$ low.** The production value $1.401 R_{\text{J}}$ sits just below the bracket $[1.472, 1.582]$. Given the isothermal- T_{eq} assumption this is within the method’s spread, but it suggests the $1\text{-}\mu\text{bar}$ level may lie slightly higher than assumed; the effect on $\dot{M}$ is bounded by the Salz base-sensitivity results ($\lesssim 10\%$ -level).
- **WASP-121 b — the atomic base is verified.** At $T_{\text{eq}} = 2358 \text{ K}$ the equilibrium base is already atomic ($q_{\text{H}_2} = 0.005$, $q_{\text{H}} = 0.92$, $\mu = 1.24$): for this ultra-hot Jupiter EXHALE’s atomic assumption holds *even in equilibrium*, with no appeal to photochemistry. The production base radius $2.2075 R_{\text{J}}$ (derived earlier from the Huang et al. 2023 Table-3 $\log g$ anchoring) lies $\sim 4\text{--}10\%$ above the isothermal- T_{eq} bracket $[2.010, 2.133]$ — consistent with a dayside lower atmosphere hotter than the redistribution-averaged T_{eq} , which extends the column. Note the deep layers are still molecular ($q_{\text{H}_2}(1 \text{ bar}) \approx 0.83$), so μ varies from 2.25 to 1.24 along this column — the equilibrium

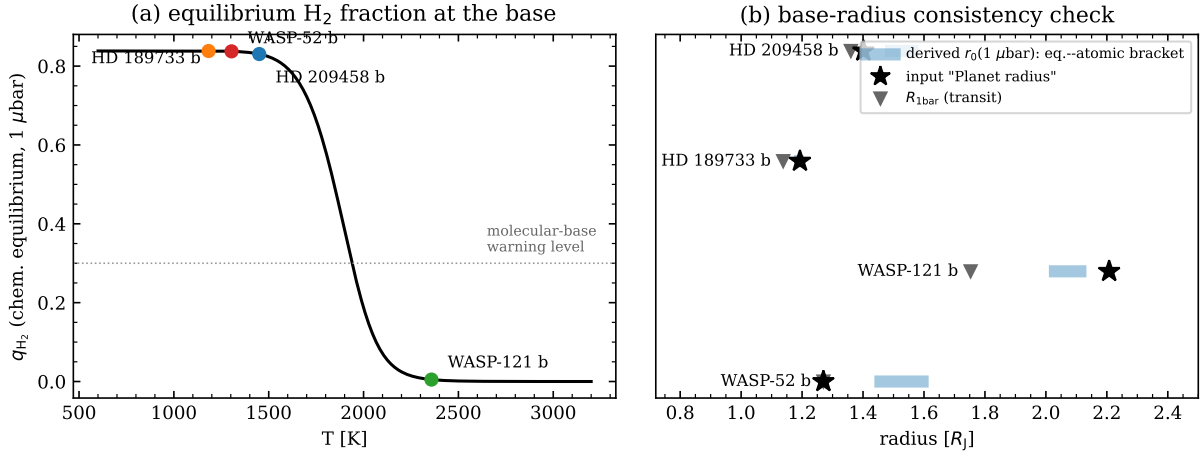


Figure 3: Tier-1 analytic column applied to the four planets. (a) The chemical-equilibrium H_2 mixing ratio at 1 μ bar (Eq. 2) as a function of temperature: the three $T_{\text{eq}} \lesssim 1500$ K planets sit on the fully molecular plateau, while WASP-121 b ($T_{\text{eq}} = 2358$ K) is atomic even in equilibrium. (b) Derived 1- μ bar base radius (equilibrium-atomic bracket, shaded) versus the production “Planet radius” (star) and the transit radius (triangle): HD 189733 b falls inside its bracket, HD 209458 b sits $\sim 5\%$ below, WASP-121 b lies just above (dayside-hot lower atmosphere), and WASP-52 b’s transit-radius shortcut puts the base $0.17\text{--}0.35 R_J$ too deep.

track is *shorter* than the atomic bracket here because of its heavier deep-layer μ .

- WASP-52 b — misplaced base, impact quantified.** The production runs use the transit radius itself ($1.27 R_J$) as the 1- μ bar base — the common shortcut — but the column places that level at $1.44\text{--}1.62 R_J$, i.e. the base is $0.17\text{--}0.35 R_J$ (13–27%) too deep. *Recomputation* (the fxuv1p0_he98 configuration rerun Newton-converged with the corrected equilibrium-column base $r_0 = 1.437 R_J$, all else identical): $\log \dot{M} \ 11.97 \rightarrow 12.14$ ($\times 1.5$), but the He 10830 observable barely moves — line-center absorption $27.11 \rightarrow 27.56\%$ ($\times 1.02$) and equivalent width $0.713 \rightarrow 0.722 \text{ \AA}$ ($\times 1.01$). The line-forming effective radius is set by the $\tau=1$ surface of the extended wind, which stays nearly fixed in *physical* units (the larger, less-bound base raises $\dot{M}$, but the He 2^3S column at a given physical radius is almost unchanged: peak $n_{2^3\text{S}} \ 150.5 \rightarrow 147.6 \text{ cm}^{-3}$). Conclusion: the base misplacement biases the *mass-loss rate* at the $\times 1.5$ level but appears not to bias the He 10830 scan conclusions materially. Figure 4 shows the two line profiles. Also note this run family’s low He/H ($= 0.0204$, the “He98” abundance scan), visible in the light μ_{base} .
- Common pattern.** All three $T_{\text{eq}} \lesssim 1500$ K planets have strongly molecular *equilibrium* bases ($q_{H_2} \approx 0.83$, the fully-molecular limit of the fit) — quantifying that their atomic bases rest on *photochemical* dissociation (Moses et al. 2011; Koskinen et al. 2013a) rather than thermal, which is precisely what the Tier-3 VULCAN upgrade would supply from first principles. The Guillot column is systematically slightly more compact (cooler skin above the photosphere) — a $\lesssim 0.5\%$ radius effect at fixed assumptions.

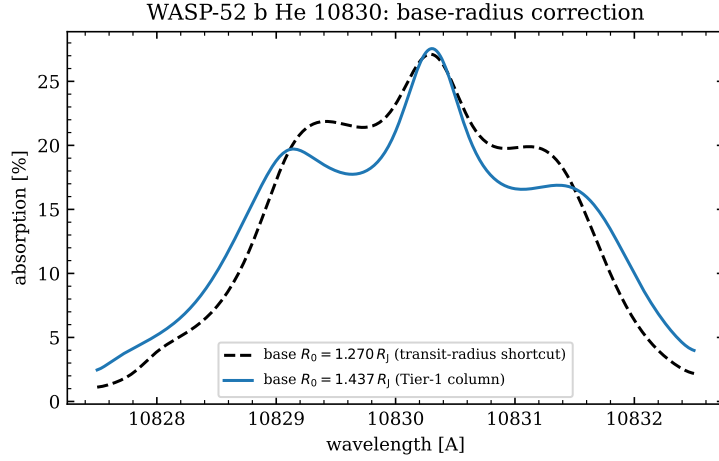


Figure 4: WASP-52 b He 10830 transmission (instrument+rotation convolved) for the old transit-radius-shortcut base ($R_0 = 1.270 R_J$, dashed) and the corrected Tier-1 column base ($R_0 = 1.437 R_J$, solid), both Newton-converged with the same binary. The line core (27.11 vs. 27.55%) and equivalent width (0.713 vs. 0.722 Å) are essentially unchanged — the $\tau=1$ line-forming surface of the extended wind is fixed in physical units — with only a mild reshaping of the shoulders; the base misplacement instead biases $\dot{M}$ ($\log \dot{M} 11.97 \rightarrow 12.14$, $\times 1.5$).

6. Using and connecting the results

6.1 Recommended workflow for each planet

1. **Column check (always, free).** Add Lower column: <transit radius> to input.inp (or run run_lower.py). Read off: does the current “Planet radius” fall inside the [chem.eq., atomic] bracket? Is q_{H_2} (base) small?
2. **Decide the tier.** $q_{\text{H}_2} \lesssim 0.1$ (e.g. WASP-121 b): the atomic model is verified — proceed as usual. $q_{\text{H}_2} \sim 0.8$ with $T_{\text{eq}} \sim 1000\text{--}2000$ K (typical hot Jupiters): the atomic base rests on photochemistry — proceed, but note the assumption; a genuinely cool planet (warm Neptune / sub-Neptune) needs Tier-2 physics before its He 10830 predictions are trusted.
3. **Adopt the geometry.** If the input radius lies outside the bracket (WASP-52 b), either correct “Planet radius” by hand or copy the driver’s base.inp into the run directory so EXHALE adopts r_0 , T_0 , He/H and K_{zz} automatically (echoed at startup).
4. **Run EXHALE** with the usual two-stage convergence (+ Newton finish for quantitative $\dot{M}$), then EXHALE_transit.py for the transit spectra.
5. **(Optional) iterate** once a real lower-atmosphere model is attached (§6.3).

6.2 Where each quantity connects

quantity	produced by	consumed by / effect
$r_0(1 \mu\text{bar})$	Tier 1 / <code>base.inp</code>	grid inner radius, gravity b_0 , irradiated area; TPM continuum radius
bracket $[r_0^{\text{eq}}, r_0^{\text{atom}}]$	Tier 1	consistency check on the input radius; systematic-error bar on r_0
T_{base}	driver (T_{eq} or Guillot)	thermal BC T_0 ; v_0 , p_0 , b_0 normalization
$q_{\text{H}_2}(\text{base})$	Eq. (2)	tier-decision switch; Molecular base correction; Tier-2 initial condition
He/H (base)	<code>base.inp</code>	composition; the He_diffusion reservoir value
K_{zz}	<code>base.inp</code>	he_kzz: homopause of the He/H (and metal) diffusive separation
$\Lambda_{\text{H}_3^+}(T, n_{\text{H}_3^+}, n_{\text{H}_2})$	<code>h3p_cooling</code>	(Tier 2) new term in the cool array of the energy equation
R1–R23 rates	<code>mol_rates</code>	(Tier 2) source terms of System_HeH_mol; $\text{H}_3^+/\text{HeH}^+$ densities for the cooling and for IR-diagnostic post-processing
metal release n_X/n_{H}	(Tier 3, VULCAN)	<code>metals.inp</code> values — replaces assumed solar totals for the Mg II/Ca II/Na I transmission lines

6.3 Feedback loop (future, with a real lower model)

One-way coupling (lower $\rightarrow$ `base.inp` $\rightarrow$ EXHALE, with the Taylor-style tolerated composition discontinuity) is the default. For the two-way iteration: EXHALE already computes the attenuated stellar spectrum on its energy grid, so the *transmitted XUV flux at the base*, $F_\nu(r_0) = F_\nu e^{-\tau_\nu(r_0)}$, can be dumped and handed to the photochemical model as its top irradiation; and EXHALE’s $\dot{M}$ supplies the lower model’s upper BC (“species < 3 amu escape at the wind velocity”, exactly the Lavvas et al. 2014 prescription). Taylor et al. found ~ 3 iterations sufficient for the analogous $\text{Ly}\alpha$ loop; the same is expected here since the wind is insensitive to base details at the $\lesssim 10\%$ level (Salz et al. 2016).

6.4 Consistency cautions

- Do not double-count composition: if `base.inp` sets He/H from a lower model, the `metals.inp` abundances should be the same model’s *atomic release* values, not solar totals.
- `base.inp` overrides win over `input.inp` for T_0 , R_0 , He/H (echoed); remove the file to return to the manual values.
- The Tier-1 column is isothermal and spherical: near-RLO planets (WASP-121 b) need the Roche correction before the bracket is taken quantitatively, and dayside-hot lower atmospheres extend beyond the T_{eq} column (both visible in the WASP-121 b example above).
- The Molecular base key changes only the EOS budget; do not combine it with quantitative He 10830 claims for genuinely cool planets — that is Tier-2 physics.

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